

This tool classifies GitHub bug reports as performance bug related or not using two approaches: a Naive Bayes + TF-IDF baseline and a fine tuned DistilBERT model. Results including metrics, statistical tests, box plots and cross project generalisation are saved automatically.

Step 1 — Clone the Repository

```
git clone https://github.com/msm-5x9/ISE-CW.git  
cd ISE-CW
```

Step 2 — Install Dependencies

```
pip install torch transformers scikit-learn pandas numpy matplotlib scipy nltk
```

Step 3 — Download the Datasets Download the five CSV files from the lab GitHub repository:

<https://github.com/ideas-labo/ISE/tree/main/lab1>

Place all five CSV files in the same directory as `br_classification_full.py` (already provided).

Step 4 — Run the Tool

```
python3 br_classification_full.py
```

Step 5 — View Results All outputs are saved to the `results/` directory automatically created in the working directory:

Output	Description
<code>results_summary.csv</code>	Median, IQR, p-value, A12 per project per metric
<code>results/<project>_raw_scores.csv</code>	All 30 raw scores per run per model
<code>results/cross_project_results.csv</code>	Cross-project generalisation results
<code>results/plots/<project>_boxplots.png</code>	Box plots per project
<code>results/plots/<project>_histograms.png</code>	Histograms per project
<code>results/plots/all_projects_f1_boxplot.png</code>	Summary F1 comparison across all projects